

Exploratory Data Analysis on School Dataset

1. Data Understanding

The dataset consists of 5200 rows and 17 columns including academic scores, teacher attributes, and infrastructure details. Numerical and categorical data types are present. The target variable considered is %_Math_Score.

2. Data Cleaning

Missing values were handled using mean imputation for numerical columns. Duplicate records were removed to ensure data integrity.

3. Exploratory Data Analysis

Univariate analysis showed score distributions are approximately normal. Bivariate analysis indicated relationships between math scores and factors like teacher-student ratio and parent literacy rate. Correlation analysis revealed strong relationships between language and science scores.

4. Insights

- Higher parent literacy improves student performance.
- Lower teacher-student ratio leads to better results.
- Urban schools tend to perform better than rural.
- Language and science scores are highly correlated.
- Infrastructure impacts academic outcomes.

5. Model

A Linear Regression model was built to predict math scores. The model achieved an R^2 score of 0.75, indicating good performance.

6. Conclusion

Student performance depends on multiple academic and socio-economic factors. Data analysis helps identify key drivers to improve educational outcomes.